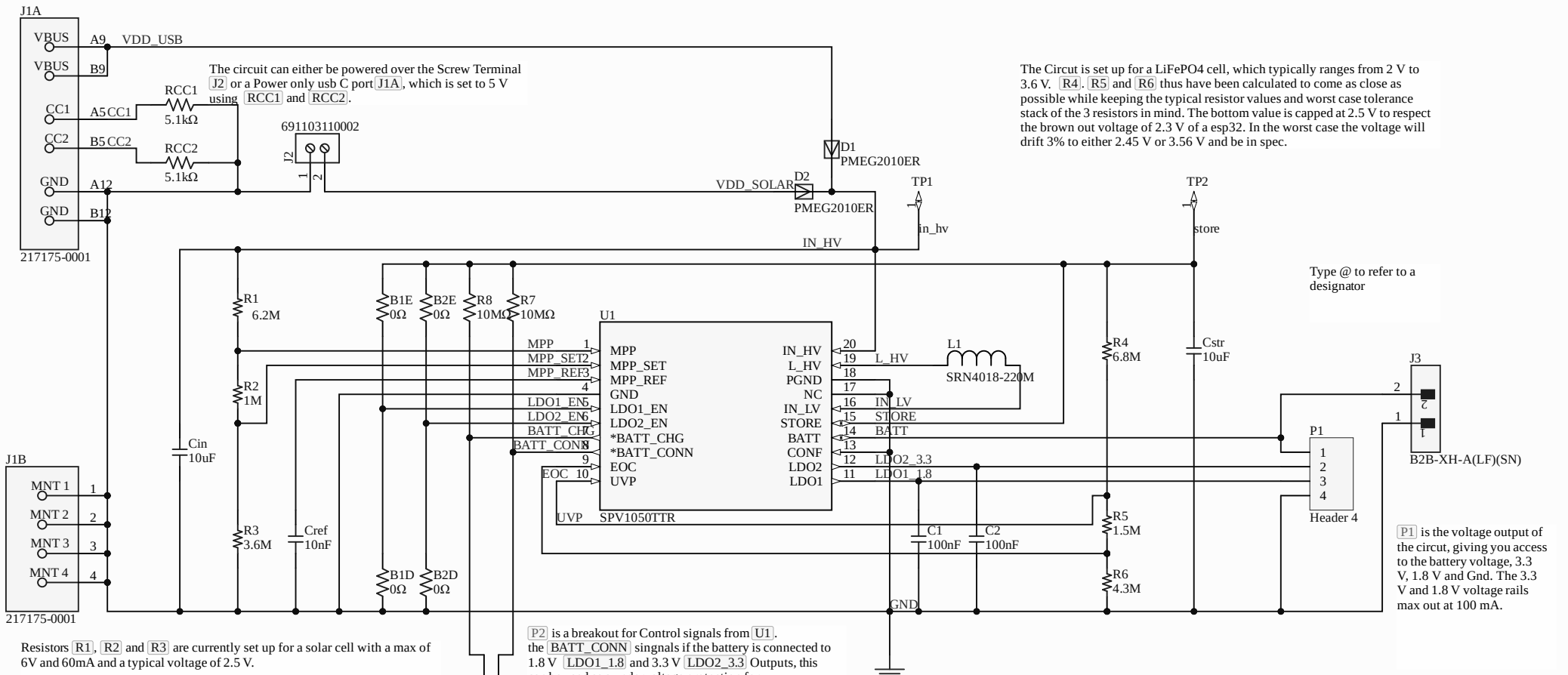


The circuit can either be powered over the Screw Terminal J2 or a Power only usb C port J1A, which is set to 5 V using RCC1 and RCC2.

The Circut is set up for a LiFePO4 cell, which typically ranges from 2 V to 3.6 V. R4. R5 and R6 thus have been calculated to come as close as possible while keeping the typical resistor values and worst case tolerance stack of the 3 resistors in mind. The bottom value is capped at 2.5 V to respect the brown out voltage of 2.3 V of a esp32. In the worst case the voltage will drift 3% to either 2.45 V or 3.56 V and be in spec.

Type @ to refer to a designator

P1 is the voltage output of the circut, giving you access to the battery voltage, 3.3 V, 1.8 V and Gnd. The 3.3 V and 1.8 V voltage rails max out at 100 mA.

Resistors R1, R2 and R3 are currently set up for a solar cell with a max of 6V and 60mA and a typical voltage of 2.5 V.

By bridging/placing a jumper on B1E enables the 3.3V output on LDO2, otherwise bridge B1D to disable it. The same goes for 1.8V on LDO1 with B1D and B1E

P2 is a breakout for Control signals from U1. the BATT_CONN singnals if the battery is connected to 1.8 V LDO1_1.8 and 3.3 V LDO2_3.3 Outputs, this can be used as a undervoltage protection for Microcontrollers etc if you are directly connected to the battery by shutting down the connected elements when BATT_CONN is pulled to gnd. BATT_CHG hpulled down when the DC-DC converter is switching, while it's released when it is not switching.

Title: LiFePO4 USB/Solar Power Board		
Size: A4	Number:	Revision: v1
Date: 10/31/2026	Sheet of 1 / 1	
File: C:\Users\..\SolarPowerBase_Sheet_1.SchDoc	Drawn By:	